

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

WEDNESDAY, 21 SEPTEMBER 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHOLOER OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST questions in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**SECTION A (Compulsory Questions)**

Q1. (a) In a C++ program, variables are declared as below:

```
int x = 2, y = 28, z = 3;
double v = 9.6, w = 2.28;
```

With the declared variables,

(i) Define the outputs of the following codes. (6 marks)

```
cout << x + y << endl;
cout << y - v << endl;
cout << v * w << endl;
cout << z % static_cast<int>(w) << endl;
```

(ii) With the declared variables above (Q1(a)), write the codes that able to COUT the exact outputs as shown in the segment below using the proper stream manipulators `#include <iomanip>`. Spacing between operands and operators are not important.

Welcome to Maths Corner (A) 2 % 28 = 2 (B) 9.6 * 3 = 28.8 (C) 28 - 2.28 + 3 = 28.7 (D) 9.6 + 2 / 2.28 = 10.48

(10 marks)

(b) Write a complete C++ program that allows user to input two integers and compare them as illustrated in Figure 1.1:

Please key in value 1: 5 Please key in value 2: 31 31 is greater than 5.
--

Please key in value 1: 100 Please key in value 2: 10 100 is greater than 10.
--

Please key in value 1: 20 Please key in value 2: 20 20 is equal to 20.
--

Figure 1.1 Sample outputs

(9 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

Q2. (a) Given $x=3$, $y=0$ and $z=1$, show step-by-step evaluations of the following expressions. Treat each of the *if* statement individually.

(i) `if ((++x > --z) || !y && !z)` (2 marks)

(ii) `if (!y && (--x - ++y - --z) && (z < 3))` (2 marks)

(b) Given the following code fragment:

```
int y, x, z = 40;
cout << "Please enter value x & y:"<<endl;
cin >> x >> y;
z *= (x++ - --y);
if (y == 4){
    if (x == 7)
        cout << z-- << "\n";
    else
        cout << z << endl;
}
cout << --z << endl;
cout << z-- + ++y << endl;
```

What are the expected outputs below,

(i) assuming $x=7$ and $y=4$? (6 marks)

(ii) assuming $x=6$ and $y=5$? (6 marks)

(c) Write a complete C++ program to read a character from user and validate whether the user's input is an **alphabet** or not by using *if...else* statement. If the input is an alphabet, determines whether the alphabet is a **vowel** or a **consonant** by using *nested if* statement. Users are allowed to re-enter the character if they wish to continue. A sample output is shown below:

Sample Output:

```
Enter an alphabet: W
"W" is a CONSONANT
Do you wish to continue?
Y

Enter an alphabet: e
"e" is a VOWEL
Do you wish to continue?
y

Enter an alphabet: @
"@ " is not an alphabet
Do you wish to continue?
N
```

(9 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

- Q3. (a) The root x of a quadratic equation e.g. $ax^2 + bx + c = 0$ can be calculated as shown in the equation below:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Equation 1: The roots of a quadratic function

The number of roots of a quadratic equation can be determined based on the discriminant ($b^2 - 4ac$) with the conditions as below:

1. $b^2 - 4ac < 0$, there are no real roots;
2. $b^2 - 4ac > 0$, there are two real roots;
3. $b^2 - 4ac$ equal to 0, there is one real root.

The following are the **THREE** functions that are related to the calculation of the root, x of a quadratic equation:

- A. **`void getInput(double &a, double &b, double &c)`**
Read a , b and c from the user and refer them to main function.
- B. **`double calculateDiscriminant(double a, double b, double c)`**
Calculate the discriminant value (using $b^2 - 4ac$) and return it to main function.
- C. **`void determineRoots(double discriminant, double a, double b, double c)`**
Calculate the root(s) (using Equation 1) based on the conditions mentioned above using the *discriminant* (obtained from *calculateDiscriminant* function) and display the values.

Answer the following questions by providing the C++ code as required below:

- (i) Function prototypes of *getInput*, *calculateDiscriminant* and *determineRoots*. (3 marks)
- (ii) Function definition of *getInput* function. (2 marks)
- (iii) Function definition of *calculateDiscriminant* function. (2 marks)
- (iv) Function definition of *determineRoots* function. (6 marks)
- (v) **MAIN** function with the respective function calls which created in Q3(a)(ii), Q3(a)(iii) and Q3(a)(iv) (following the sequences) when executed. Sample outputs are as shown below: (3 marks)

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q3. (a) (Continued)**

```
Please key in the value of a, b and c: 1 -3 4
There are no real roots.
```

Figure 3.1 Sample output: No real root when $b^2 - 4ac < 0$

```
Please key in the value of a, b and c: 2 -11 5
There are two real roots
The roots of x are 5 and 0.5.
```

Figure 3.2 Sample output: Two real roots when $b^2 - 4ac > 0$

```
Please key in the value of a, b and c: -4 12 -9
There is one real root
The root of x is 1.5.
```

Figure 3.3 Sample output: One root when $b^2 - 4ac$ equal to 0

- (b) Write a void function named ***void readInputToArray(string filename, int storeFigure[], int &totalFigure)*** that reads integers from a text file (as shown in Figure 3.4) and stored in one-dimensional array known as *storeFigure*. The total amount of integers stored in the array is referred back to the calling function in variable *totalFigure*.

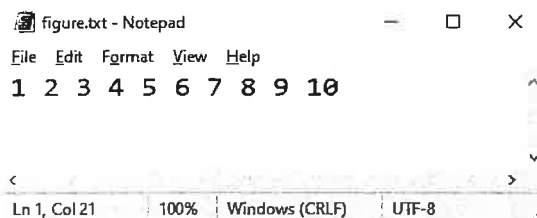


Figure 3.4 Integers store in a text file

(9 marks)

[Total : 25 marks]

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**SECTION B**

Q4. (a) The following questions are related to C++ structure.

- (i) Define a struct called ***fruitData*** that consists of two variables: ***fruitName*** as string and ***price*** as float. (2 marks)
- (ii) Declare an array of ***fruitData*** type called ***fruits*** with 20 elements. (2 marks)
- (iii) Define a struct called ***invoiceData*** that consists of four members: ***NoOfFruitItem_ordered*** as integer; ***pound_ordered*** and ***totalFruitprice*** as float; ***fruitData*** type called ***fruit*** with 20 elements. (5 marks)
- (iv) Declare an array of ***invoiceData*** type called ***customers*** with 100 elements. (2 marks)

- (b) Based on the struct ***fruitData*** defined in Q4(a)(i), write a function ***void readInput(fruitData fruits[])*** that reads the text file “fruit.txt” as shown in Figure 4.1 and stores the information into the ***fruitData*** using struct array. The total number of fruits read from the text file is stored in a global variable named ***item***. (6 marks)

```
Banana
1
Apple
2
Pear
2.5
Orange
1.5
Papaya
1.4
```

Figure 4.1 fruit.txt

The information is arranged using the structure as below:

```
[fruitName]
[price]
```

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q4. (Continued)**

- (c) Implement a function **void fruitPurchase(invoiceData customers[])** that allows user to purchase fruits by key in the fruit name and the weight of each fruit in pounds. Display and store the output to “customersHistory.txt” (Figure 4.2) as shown in the sample output below:

Sample Output:

```

Welcome to the fruit market

Banana
Price = RM 1 per pound
Apple
Price = RM 2 per pound
Pear
Price = RM 2.5 per pound
Orange
Price = RM 1.5 per pound
Papaya
Price = RM 1.4 per pound

Enter the fruit name that customer would like to purchase:
Apple
How many pounds?
3.5
Do you wish to add other fruits?
Y
Enter the fruit name that customer would like to purchase:
Orange
How many pounds?
4.5
Do you wish to add other fruits?
Y
Enter the fruit name that customer would like to purchase:
Pear
How many pounds?
2
Do you wish to add other fruits?
N
You have ordered 3 item(s) with total price of RM 18.75.
Next Purchase?
Yes

Enter the fruit name that customer would like to purchase:
Papaya
How many pounds?
5.5
Do you wish to add other fruits?
N
You have ordered 1 item(s) with total price of RM 7.70.

Next Purchase?
No

```

cout in the **MAIN** function which stored in **fruitData**

Hint: Y = Yes; N = No.

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICESQ4. (c) (Continued)

```

Customer ID: 1
Fruit type: Apple
No. of pound = 3.50
Fruit type: Orange
No. of pound = 4.50
Fruit type: Pear
No. of pound = 2.00
Customer 1 have ordered 3 item(s) with total price of RM 18.75.
Customer ID: 2
Fruit type: Papaya
No. of pound = 5.50
Customer 2 have ordered 1 item(s) with total price of RM 7.70.

```

Figure 4.2 customersHistory.txt

(8 marks)

[Total : 25 marks]

Q5. **PlayerUnknown's Battlegrounds (PUBG)** is a very popular online multiplayer battle royale game published in 2017 by PUBG Corporation. It is a sole survivor game which players are placed in a game area to battle with each other and the last player wins the round.

- (a) Using C++ code to define a struct called **PUBG** that consists of 4 types of variables: **username of PUBG** as string, **total games played** as integer, **hour's games played** as double and **current online status (yes or no)** as boolean.

(6 marks)

- (b) Assuming struct array **PUBG PUBGAcc[]** and **int totalPlayer** are declared and initialised at the **MAIN** function as shown below:

```

int main(){
    PUBG PUBGAcc[100]; //struct array
    int totalPlayer = 0; //Record the total player of PUBG
    int answer = 1; //To obtain information of multiple users

    while (answer == 1) {
        readUser(PUBGAcc, totalPlayer);

        cout << "Continue? 1 - Yes, 2 - No: ";
        cin >> answer;
        cin.ignore();
    }

    cout << "Total games played by users is "
        << totalGamesPlayedByAllUser(PUBGAcc, totalPlayer)
        << ". " << endl;
    checkCurrentOnlineStatus(PUBGAcc, totalPlayer);
    return 0;
}

```


UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES**Q5. (b) (Continued)**

- (i) Based on the struct **PUBG** defined in Q5(a), write function definition of **void readUser(PUBG PUBGAcc[], int &totalPlayer)** (as shown in the **MAIN** function) that able to store the information of multiple users (key in by user in this case) into **PUBG** using struct array and also update the total number of players. (6 marks)
- (ii) Write function definition of **int totalGamesPlayedAllUser(PUBG PUBGAcc[], int totalPlayer)** (as shown in the **MAIN** function) that calculates the total games played by all the users that stored in **PUBGAcc** as shown in Figure 5.1. (6 marks)
- (iii) Write function definition of **void checkCurrentOnlineStatus(PUBG PUBGAcc[], int totalPlayer)** (as shown in the **MAIN** function) that determines the players with the current status show online. The function also calculates the total current online players and displays the output as shown in Figure 5.1. (7 marks)

```

Create User Account of PUBG

Username: Tan Joi San
Total Games Played: 10
Hour's Games Played: 13.5
Current online status: 1
Continue? 1 - Yes, 2 - No: 1

Username: Jasmina Khaw Yen Min
Total Games Played: 99
Hour's Games Played: 110
Current online status: 0
Continue? 1 - Yes, 2 - No: 1

Username: Richard Rich
Total Games Played: 5
Hour's Games Played: 6
Current online status: 1
Continue? 1 - Yes, 2 - No: 2

Total games played by users is 114.

Current online player:
Tan Joi San
Richard Rich
Total of current online player is 2.

```

Figure 5.1 Example output of the program

[Total : 25 marks]

